

Sunilkumar Pathipati

Lead Engineer | Python · AI · Forward Deployed Engineering · AUTOSAR

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Professional Summary

Lead Engineer with 9+ years in automotive software, embedded systems, and applied AI. Works in Python and C on AUTOSAR programs, machine-learning workflows, and Agile delivery. Safety-critical experience includes anti-lock braking software for eBikes and two-wheelers in ISO 26262 environments. Currently building Python automation and on-prem LLM tooling for infotainment test, alongside software integration, verification and validation, release management, and engineering productivity. Pursuing a PG Certificate in Forward Deployed AI Engineering (Futureense × IIT Roorkee).

Technical Skills

- **Languages:** Python, C, Embedded C, Bash, MATLAB
- **AI & Machine Learning:** Artificial Intelligence, Machine Learning, LLMs, llama.cpp, Data Analytics
- **Automotive:** AUTOSAR, ISO 26262, Android Infotainment, ABS Systems, Embedded Systems
- **Tools:** Git, GitHub, GitHub Actions, JFrog Artifactory, Linux, Simulink
- **Core:** Test Automation Frameworks, Software Validation, System Integration, Release Management
- **Methods:** Agile Development, CI/CD Workflows, Software Development Life Cycle, Verification & Validation
- **Spoken languages:** English, Telugu, Hindi, Kannada, Tamil

Professional Experience

Lead Engineer

BlueBinaries Engineering and Solutions Pvt Ltd (Client: Polestar) | Bengaluru, Karnataka, India | September 2025 -- Present

- Lead Python automation and local llama.cpp LLM tooling for Polestar infotainment test and validation.
- Build AI tools for test generation, log analysis, defect investigation, knowledge retrieval, and engineering productivity.
- Own GitHub Actions workflows and Git-based development processes that streamline test, build, and release work.
- Use Python, Bash, and JFrog Artifactory to raise software quality and shorten validation cycles.
- Deploy and operate on-premises LLM tooling so automotive test environments stay off the public cloud.

Lead Engineer

Bosch Global Software Technologies | Bengaluru, India · Reutlingen, Germany | January 2024 -- August 2025

- Led automotive software development and validation for safety-critical embedded systems, including ISO 26262-compliant processes.
- Managed software integration, release management, verification, and validation across global programs.
- Led automation initiatives and engineering workflow improvements with international teams in Germany and India.
- Contributed to algorithm development, application development, and software release activities for automotive systems.

Specialist

Bosch Global Software Technologies | Bengaluru, Karnataka, India | January 2023 -- December 2023

- Developed and validated automotive software components for eBike and two-wheeler braking systems.
- Designed automation solutions and test frameworks to improve validation efficiency and software quality.

Senior Software Engineer

Bosch Global Software Technologies | Bengaluru, India · Reutlingen, Germany | December 2021 -- January 2023

- Worked on embedded software development, testing, debugging, and integration for automotive applications.
- Supported MATLAB/Simulink-based algorithm development and validation activities.
- Collaborated with international engineering teams in Germany on validation, automation, and software quality.

Software Engineer

Bosch Global Software Technologies | Bengaluru, Karnataka, India | July 2019 -- December 2021

- Developed embedded automotive software and supported testing activities for safety-critical systems.
- Worked on software verification, validation, debugging, and release support.

Software Engineer

HCL Technologies | Bengaluru, Karnataka, India | March 2017 -- July 2019

- Developed and tested embedded software solutions for automotive clients.
- Supported system integration, validation, debugging, and software testing activities.
- Worked with Linux-based development environments and automotive software tools.

Projects

LLM JUnit4 Test Generator | Python | Freelance

<https://github.com/psunil0001-sys/LLM-Junit4-Unittest-Generator>

Local LLM pipeline that drafts and repairs Kotlin/JUnit4 unit tests for Android Gradle modules. A Google ADK planner, coder, and fixer loop runs Gradle and Kover (optional JaCoCo) and edits only generated tests until selected tasks pass, using llama.cpp on-prem or a remote OpenAI-compatible vLLM endpoint.

RevOps Research Desk | Python | Freelance

<https://github.com/psunil0001-sys/revops>

Streamlit workspace for mutual funds, ULIPs, and NPS holdings from mfapi.in and local CSVs. A manager coordinates five analysts (fundamentals, sentiment, news, technical, policy), stores notes in ChromaDB, and produces 60-day NAV scenarios with bootstrap resampling and a local LLM.

CalorieTrack | Dart | Freelance

<https://github.com/psunil0001-sys/calorietrack>

Cross-platform Flutter calorie tracker with a layered structure (core, presentation, routes, theme, widgets), typed navigation, and a light/dark theme system for a consistent mobile UI.

Airflow TaskFlow ETL | Python | Freelance

<https://github.com/psunil0001-sys/DAG-example>

Astronomer / Apache Airflow example of TaskFlow ETL: pull live astronaut data, then process each result with dynamic task mapping for local orchestration.

Recruiter Portfolio | TypeScript | Project

<https://github.com/psunil0001-sys/portfolio>

Recruiter-facing site and ATS resume pipeline: React and Vite on GitHub Pages, with a Python helper that rebuilds live project cards and a single-column resume from LinkedIn PDF exports and public GitHub repositories.

Education

Bachelor of Technology in Electronics and Communications Engineering

Ramireddy Subbaramireddy Engineering College | Andhra Pradesh, India | 2012 -- 2016

Certifications

- PG Certificate in Forward Deployed AI Engineering (Futureense × IIT Roorkee) — in progress, expected Oct 2026
- Certified Data Scientist
- Data Science Foundation
- Deep Learning with Keras and TensorFlow